UNIVERSIDAD DE GUAYAQUIL

FACULTAD DE CIENCIAS, MATEMATICAS Y FISICAS

CARRERA DE SOFTWARE

MATERIA:

GESTION DE LA CONFIGURACION DEL SOFTWARE

TEMA DE ACTIVIDAD:

PRACTICA INTEGRADORA SEMANA 1 A SEMANA 5

DOCENTE:

ING. ALVAREZ SOLIS FRANCISCO XAVIER

ESTUDIANTE:

TENESACA PINCAY CINTHIA MARIA

CICLO ESCOLAR:

2025 – 2026 CICLO II

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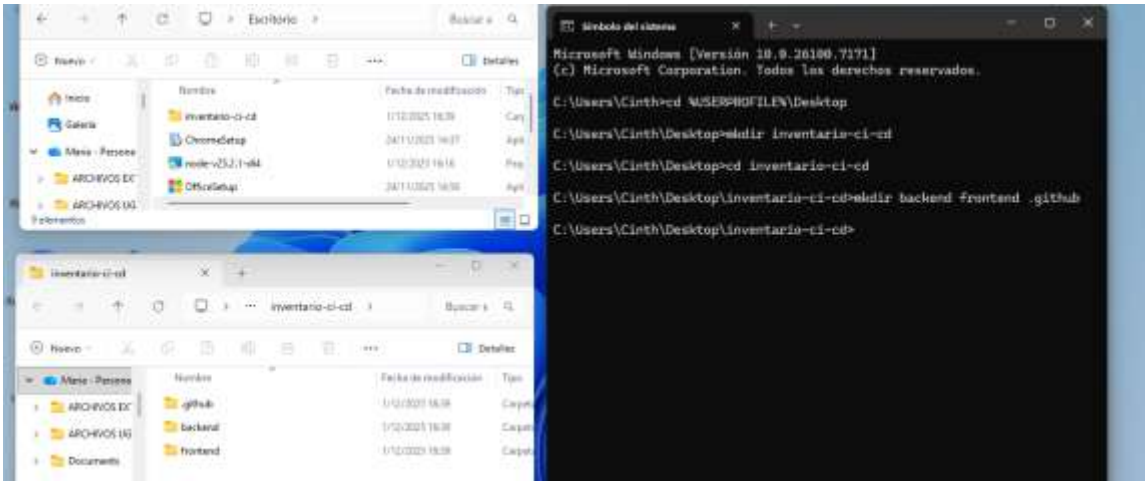
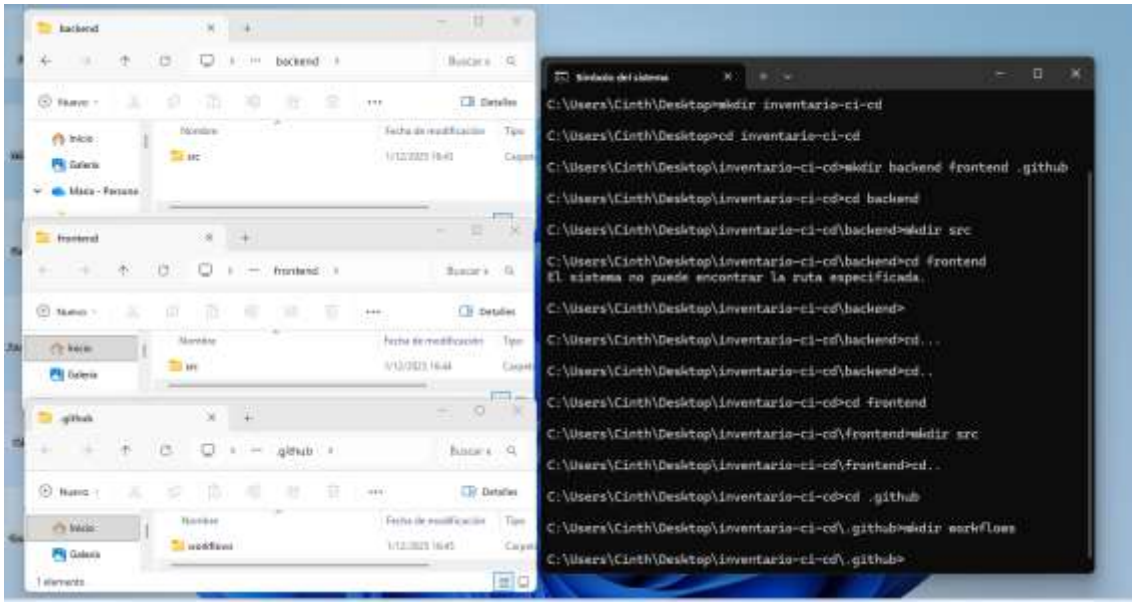
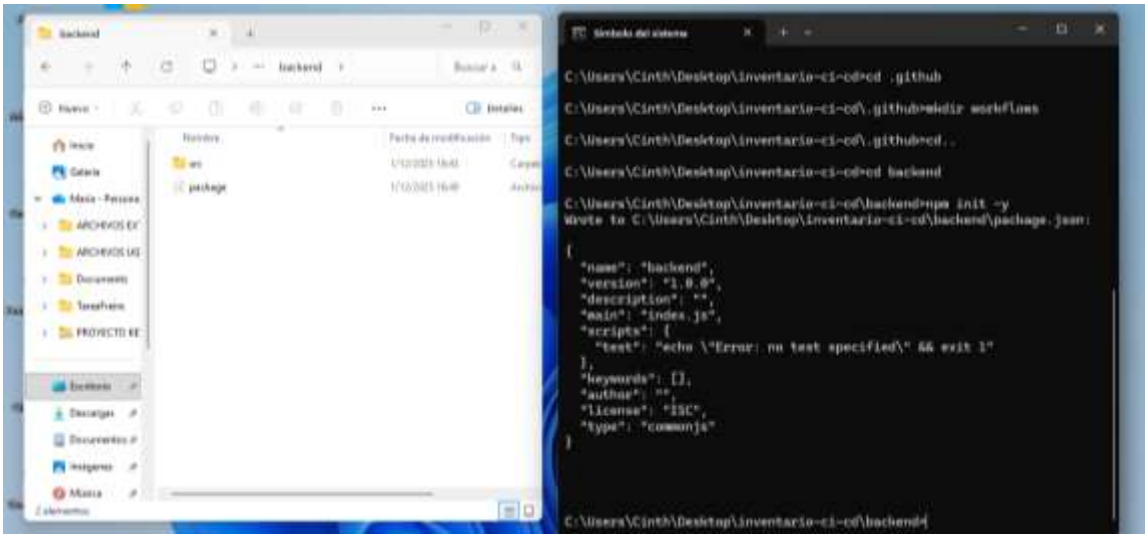
PRACTICA CI/CD

BACKEND

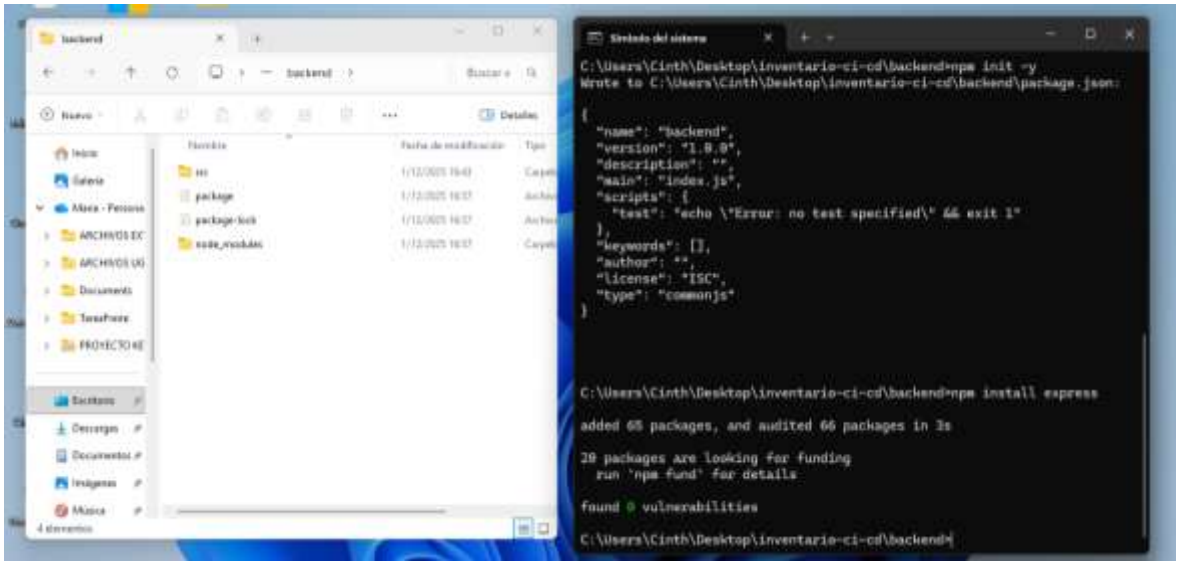




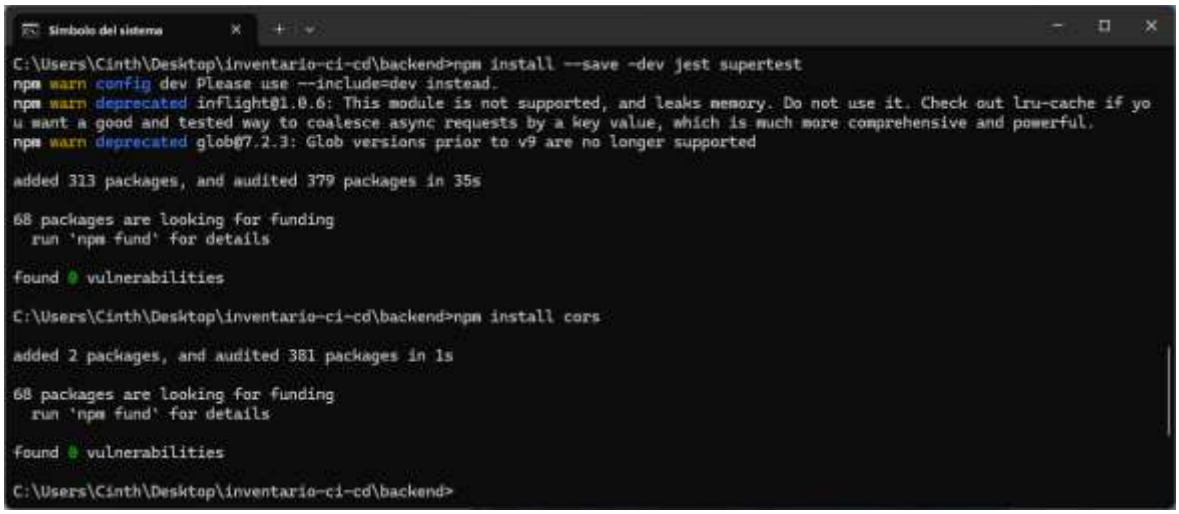
 # Crear el archivo package.json con valores por defecto:



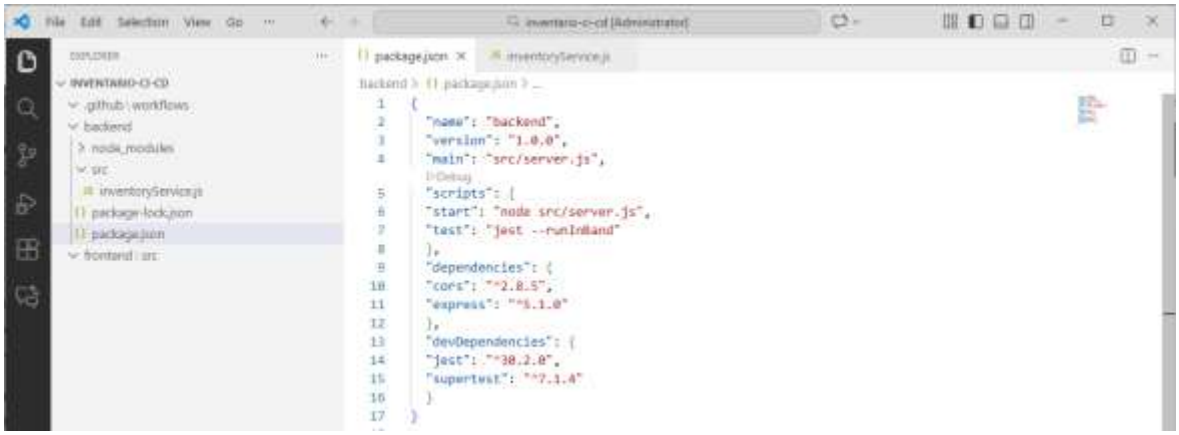
 # Dependencia de Runtime



 # Dependencias de desarrollo (pruebas)



Archivo: package.json



Archivo: src/inventoryService.js

```javascript
const productos = [];
let ultimoId = 0;

function crearProducto(data){
        if(!data.sku || !data.nombre){
        throw new Error('SKU y nombre son obligatorios');
    }
    const stockInicial = data.stock ?? 0;

    const nuevo = {
        id: ++ultimoId,
        sku: String(data.sku),
        nombre: String(data.nombre),
        stock: Number(stockInicial)
    };
    productos.push(nuevo);
    return nuevo;
}
function listarProductos(){
    return [...productos];
}
module.exports = {
    crearProducto,
    listarProductos
};
```

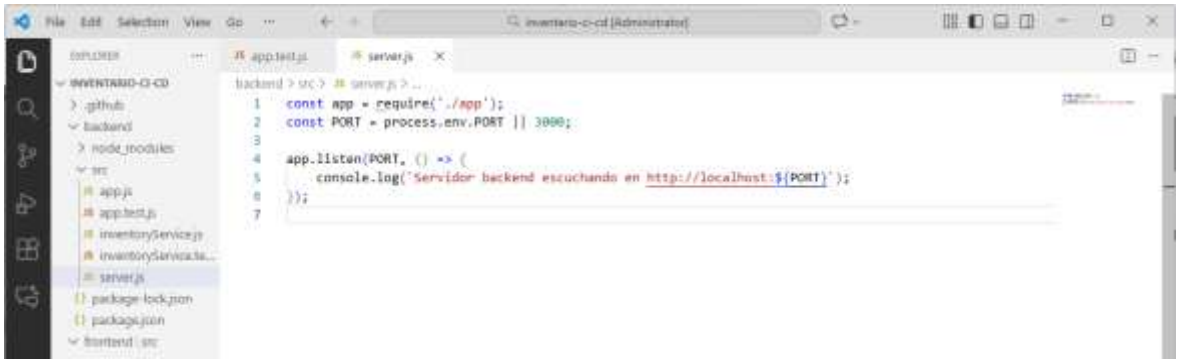
Archivo: src/app.js

```javascript
const express = require('express');
const cors = require('cors'); // - IMPORTANTE
const { crearProducto, listarProductos } = require('./inventoryService');
const app = express();

app.use(cors()); // - HABILITA CORS
app.use(express.json()); // - NECESARIO PARA PROCESAR JSON
// Listar productos
app.get('/api/productos', (req, res) => {
    const items = listarProductos();
    res.json({ data: items });
});
// Crear producto
app.post('/api/productos', (req, res) => {
    try {
        const nuevo = crearProducto(req.body);
        res.status(201).json({ data: nuevo });
    } catch (err) {
        res.status(400).json({
            error: 'VALIDATION_ERROR',
            message: err.message
        });
    }
});
module.exports = app;
```

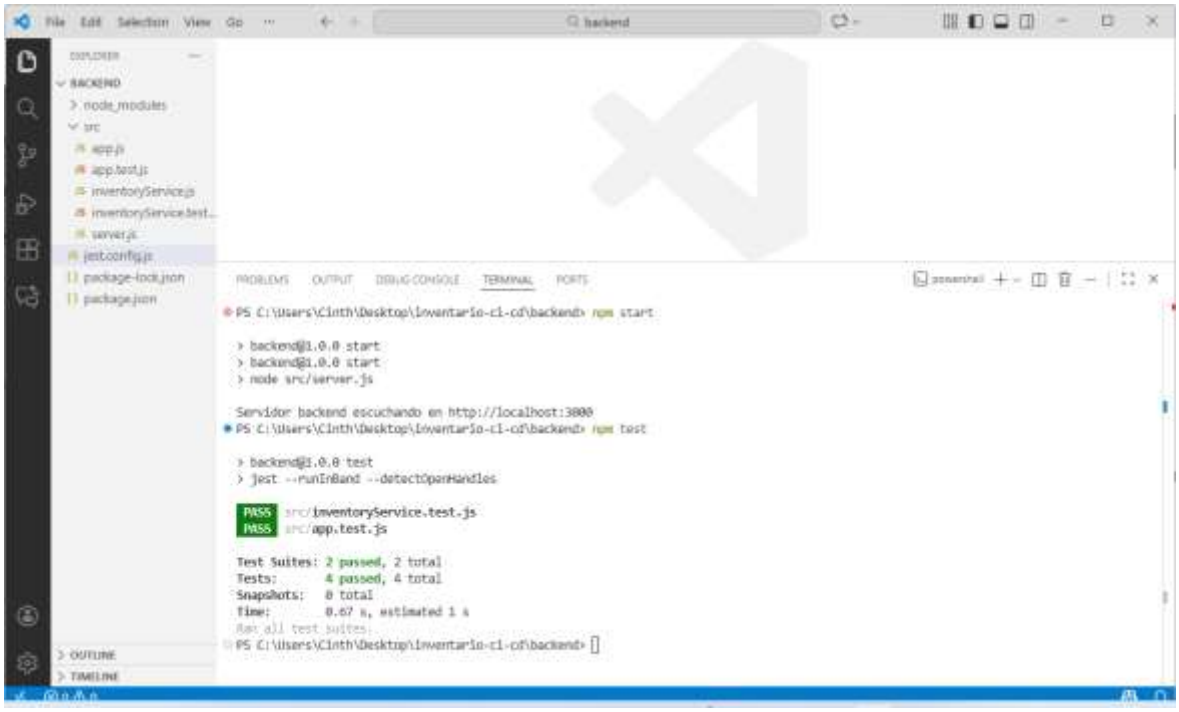
Archivo: src/inventoryService.test.js

```javascript
const { captureRejectionSymbol } = require('supertest/lib/test');
const { crearProducto, listarProductos } = require('./inventoryService');

beforeEach(() => {
    jest.resetModules();
});
test('crea un producto válido con stock inicial 0 si no se define', () => {
    const { crearProducto } = require('./inventoryService');
    const prod = crearProducto({
        sku: 'A-001',
        nombre: 'Cable HDMI'
    });
    expect(prod.id).toBe(1);
    expect(prod.sku).toBe('A-001');
    expect(prod.nombre).toBe('Cable HDMI');
    expect(prod.stock).toBe(0);
});

test('lanza error si falta sku o nombre', () => {
    const { crearProducto } = require('./inventoryService');
    expect(() => crearProducto({ sku: 'A-002' }))
        .toThrow('SKU y nombre son obligatorios');
    expect(() => crearProducto({ sku: 'Mose' }))
        .toThrow('SKU y nombre son obligatorios');
});
```

app.test.js



Verificar El Backend

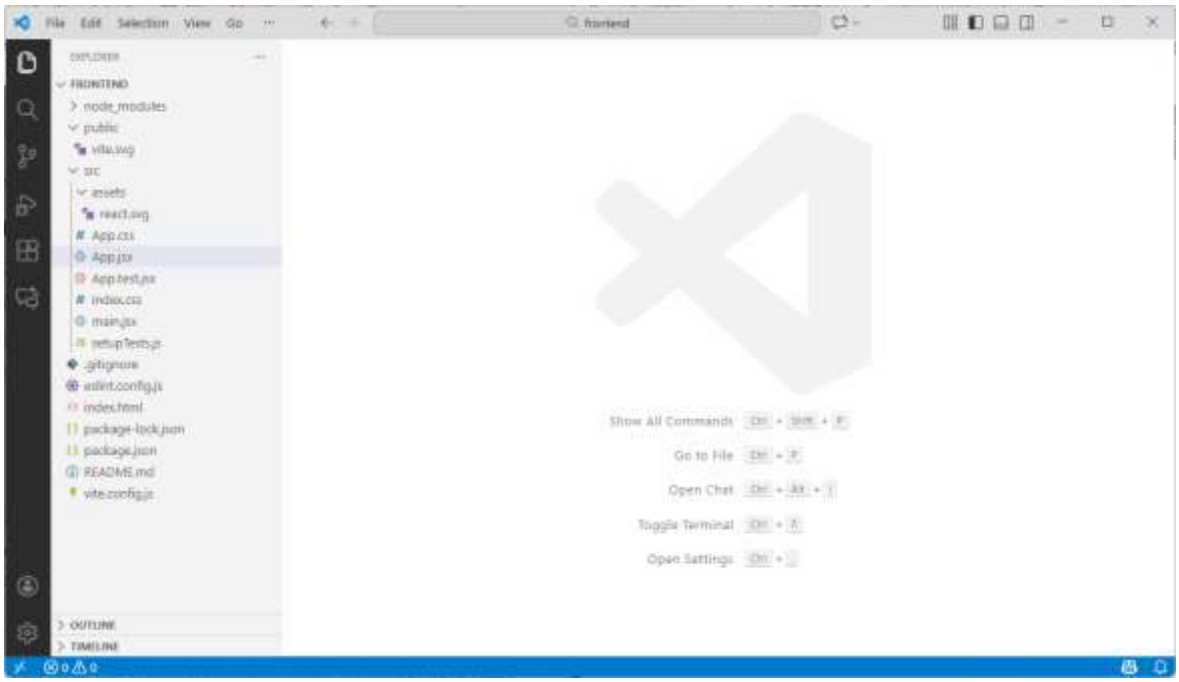


FRONTEND

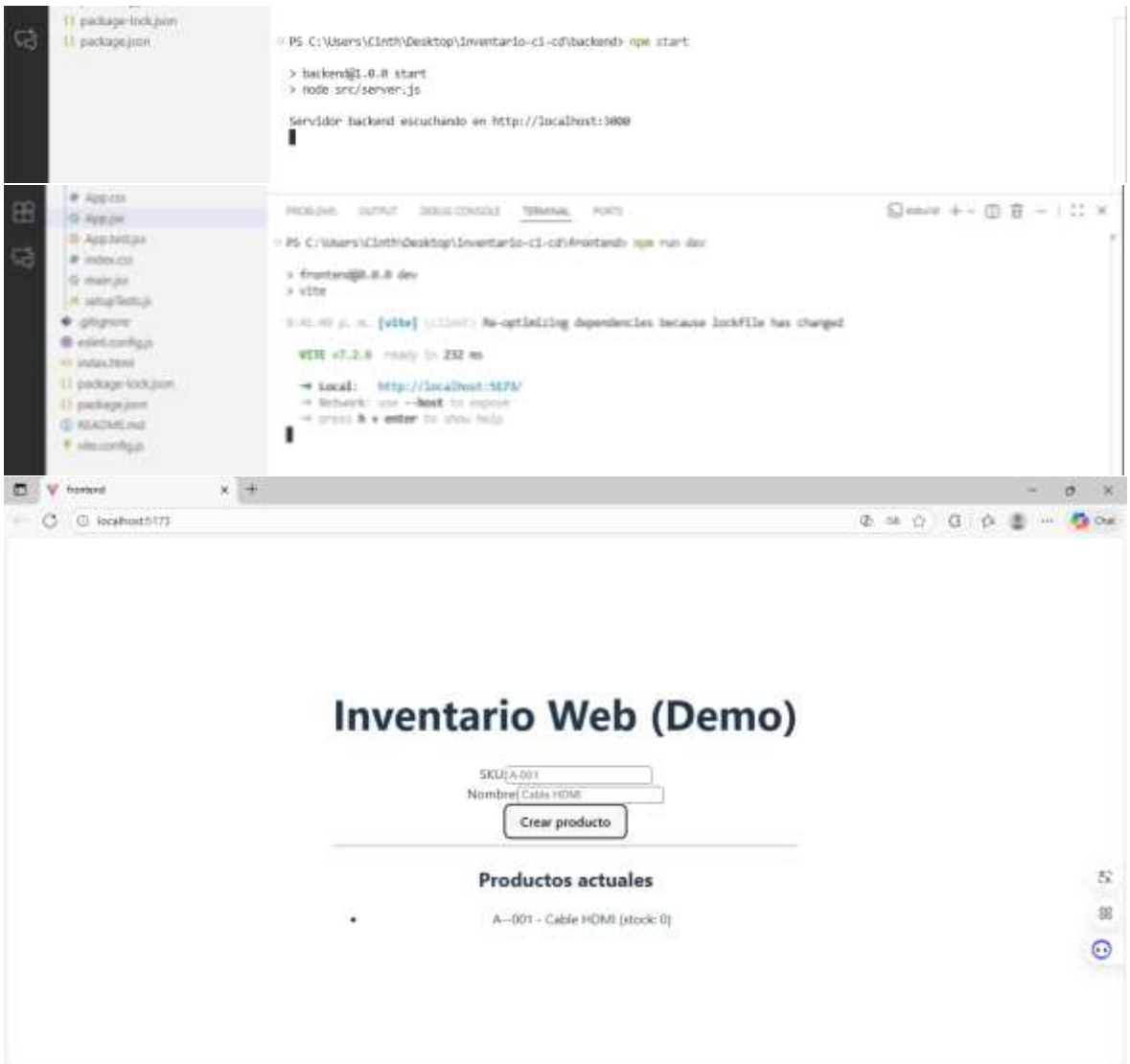
```
npm install
Microsoft Windows [Versión 10.0.26100.7171]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\Cinth>cd C:\Users\Cinth\Desktop\inventario-ci-cd

C:\Users\Cinth\Desktop\inventario-ci-cd>cd frontend

C:\Users\Cinth\Desktop\inventario-ci-cd\frontend>npm create vite@latest . -- --template react
Need to install the following packages:
create-vite@8.2.0
Ok to proceed? (y) y


> npx
> create-vite . --template react

|
◇  Current directory is not empty. Please choose how to proceed:
|  Remove existing files and continue
|
◇  Use rolldown-vite (Experimental)?:
|  No
|
◇  Install with npm and start now?
|  Yes
|
◇  Scaffolding project in C:\Users\Cinth\Desktop\inventario-ci-cd\frontend...
|
◇  Installing dependencies with npm...
└
```

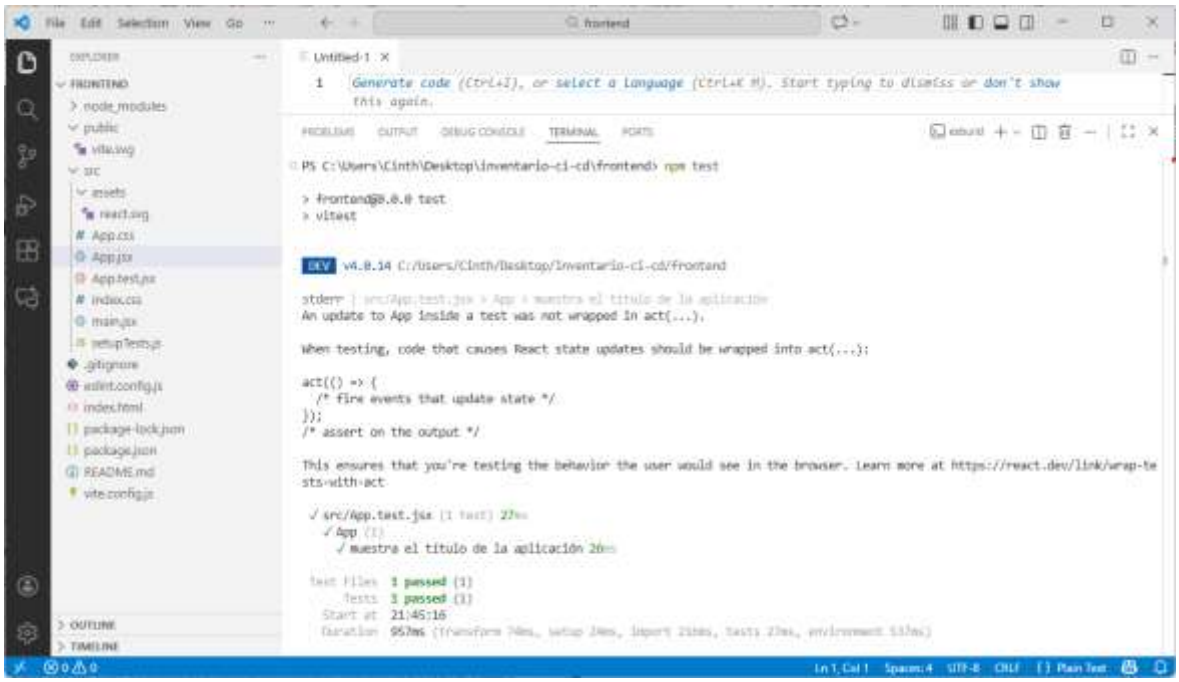
```
Símbolo del sistema

  VITE v7.2.6  ready in 2141 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
¿Desea terminar el trabajo por lotes (S/N)?
^C
C:\Users\Cinth\Desktop\inventario-ci-cd\frontend>npm install

up to date, audited 158 packages in 730ms

33 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Cinth\Desktop\inventario-ci-cd\frontend>npm install -D vitest @testing-library/react @testing-library/user-even
t jsdom

added 82 packages, and audited 240 packages in 13s

51 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Cinth\Desktop\inventario-ci-cd\frontend>
```

```
package.json
{
  "name": "frontend",
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview",
    "test": "vitest"
  },
  "dependencies": {
    "react": "^19.2.0",
    "react-dom": "^19.2.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.39.1",
    "@testing-library/jest-dom": "^6.9.1",
    "@testing-library/react": "^16.3.0",
    "@testing-library/user-event": "^14.6.1",
    "@types/react": "^19.2.5",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^5.1.1",
    "eslint": "^9.39.1",
    "eslint-plugin-react-hooks": "^7.0.1",
    "eslint-plugin-react-refresh": "^0.4.24",
    "globals": "^16.5.0",
    "jsdom": "^27.2.0",
    "vite": "^7.2.4",
    "vitest": "^4.0.14"
```



Ejecución del frontend



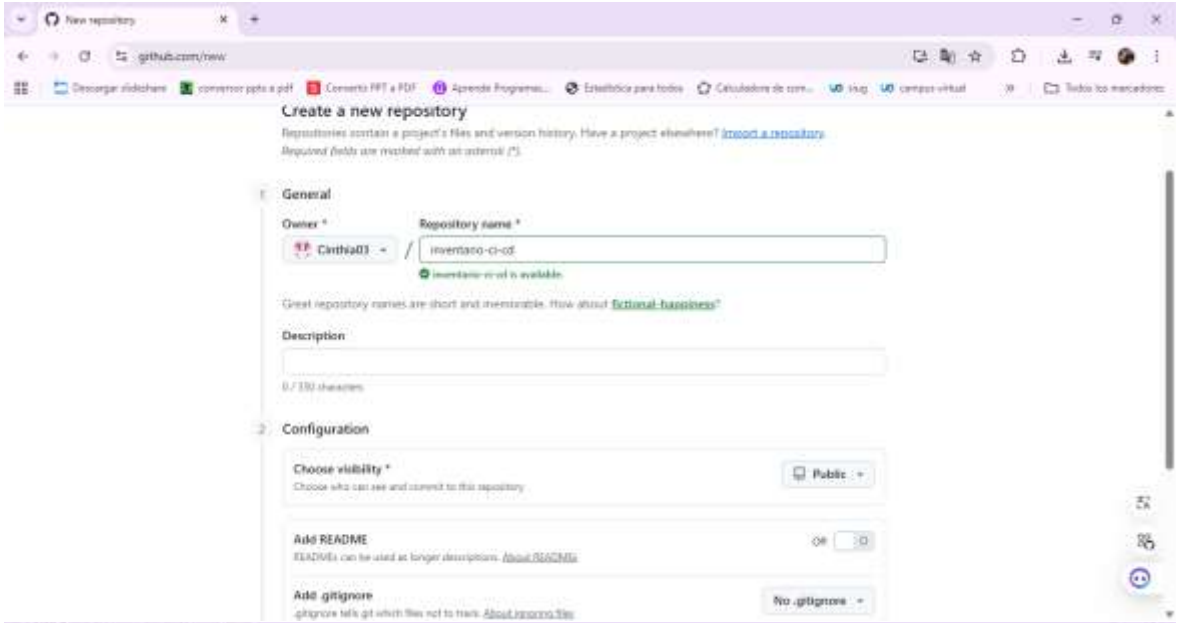

File Edit Selection View Go
frontend
EXPLORER
FRONTEND
node_modules
public
vite.svg
src
assets
react.svg
App.css
App.jsx
App.test.jsx
index.css
main.jsx
setupTests.js
.gitignore
eslint.config.js
index.html
package-lock.json
package.json
README.md
vite.config.js
OUTLINE
TIMELINE
Untitled-1
Generate code (Ctrl+I), or select a Language (Ctrl+K M). Start typing to dismiss or don't show this again.
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Cinth\Desktop\inventario-ci-cd\frontend> npm test
> frontend@0.0.0 test
> vitest
DEV v4.0.14 C:/Users/Cinth/Desktop/inventario-ci-cd/frontend
stderr | src/App.test.jsx > App > muestra el titulo de la aplicación
An update to App inside a test was not wrapped in act(...).
When testing, code that causes React state updates should be wrapped into act(...):
act(() => {
/* fire events that update state */
});
/* assert on the output */
This ensures that you're testing the behavior the user would see in the browser. Learn more at https://react.dev/link/wrap-tests-with-act
✓ src/App.test.jsx (1 test) 27ms
✓ App (1)
✓ muestra el titulo de la aplicación 26ms
Test Files 1 passed (1)
Tests 1 passed (1)
Start at 21:45:16
Duration 957ms (transform 74ms, setup 24ms, import 216ms, tests 27ms, environment 517ms)
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Plain Text

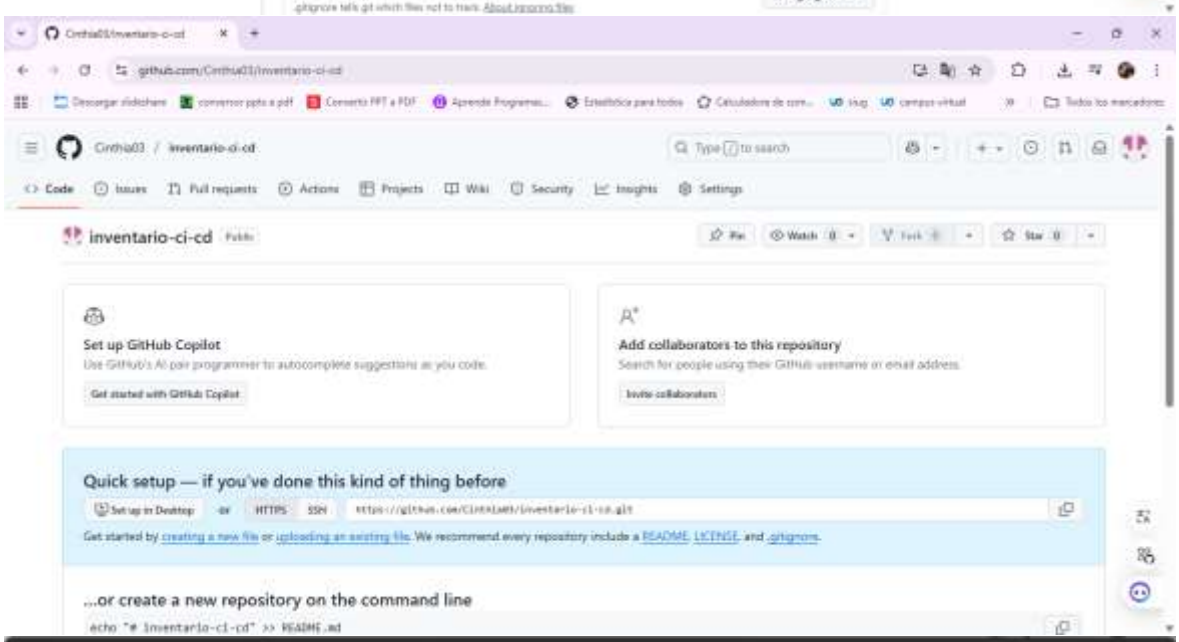
INTEGRACIÓN CONTINUA Y DESPLIEGUE CONTINUO (CI/CD)

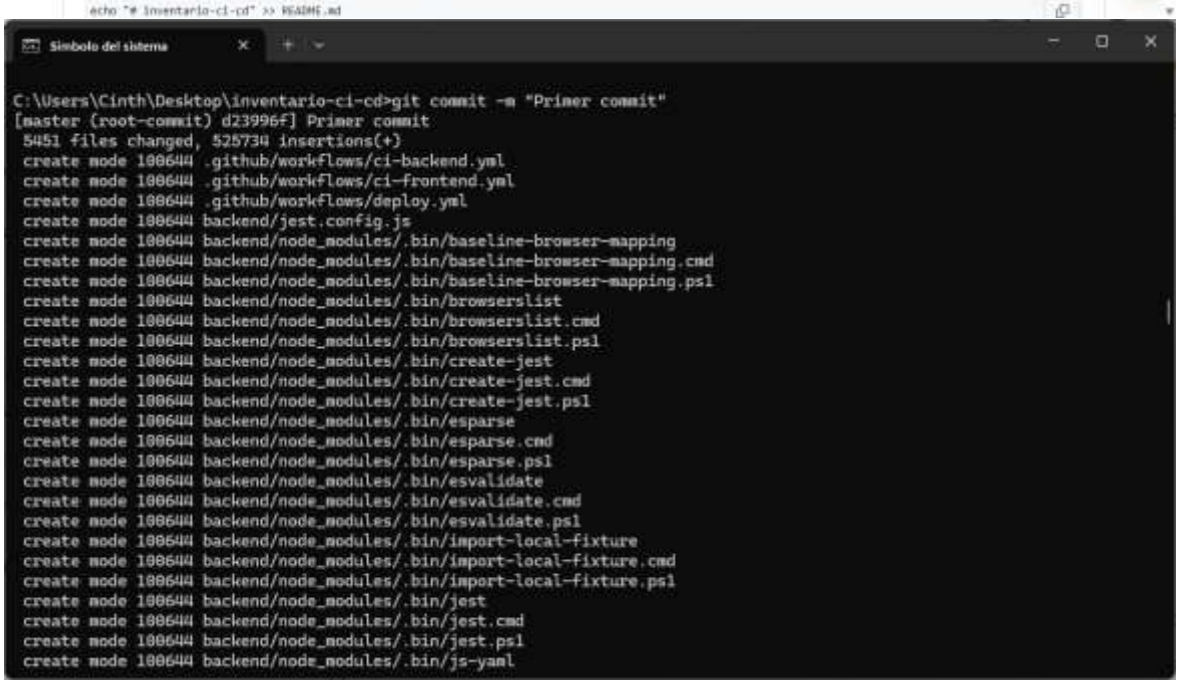
```
Símbolo del sistema
Microsoft Windows [Versión 10.0.26100.7171]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\Cinth>C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows
"C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.

C:\Users\Cinth>cd C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>git --version
git version 2.52.0.windows.1

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>git config --global user.name "CinthiaTenesaca"

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>git config --global user.email "cinthia2003maria@gmail.com"

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
```

```
Símbolo del sistema
init.defaultbranch=master
user.name=CinthiaTenesaca
user.email=cinthia2003maria@gmail.com

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>ssh-keygen -t ed25519 -C "cinthia2003maria@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\Cinth/.ssh/id_ed25519):
Created directory 'C:\\Users\\Cinth/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Cinth/.ssh/id_ed25519
Your public key has been saved in C:\Users\Cinth/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Hm9UsBiCnKulyLF/MP4BDe8qiYmNW4mvD/8tD1l1BLg cinthia2003maria@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|       ....      |
|      .  .       |
|       .. . o    |
|     .E= o o o   |
|+     O S * .    |
|.B   oo* =       |
|.+=o+ooo+ o      |
|o.+o+O.o.o       |
|..  *=++o        |
+----[SHA256]-----+

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>ssh -T git@github.com
The authenticity of host 'github.com (140.82.112.3)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
```

```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>type C:\Users\Cinth\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM2YLFiz3kYUxbY0lIbjV1s62DReJvOsZO+kaHRwjUwi cinthia2003maria@gmail.com

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>ssh -T git@github.com
Hi Cinthia03! You've successfully authenticated, but GitHub does not provide shell access.

C:\Users\Cinth\Desktop\inventario-ci-cd\.github\workflows>
```

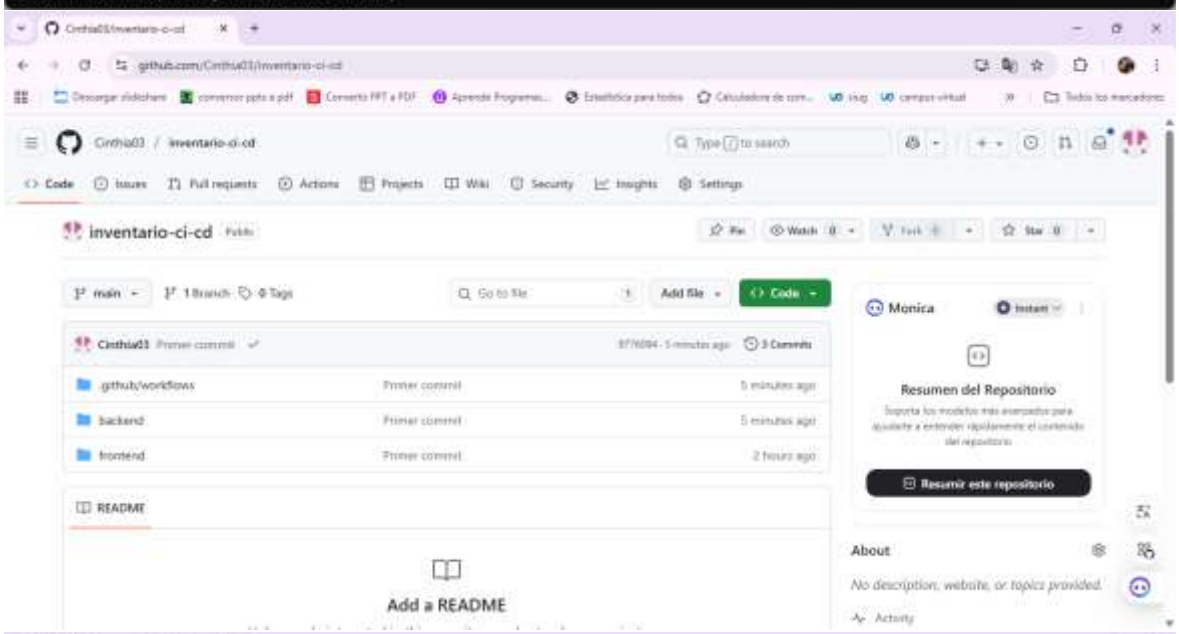


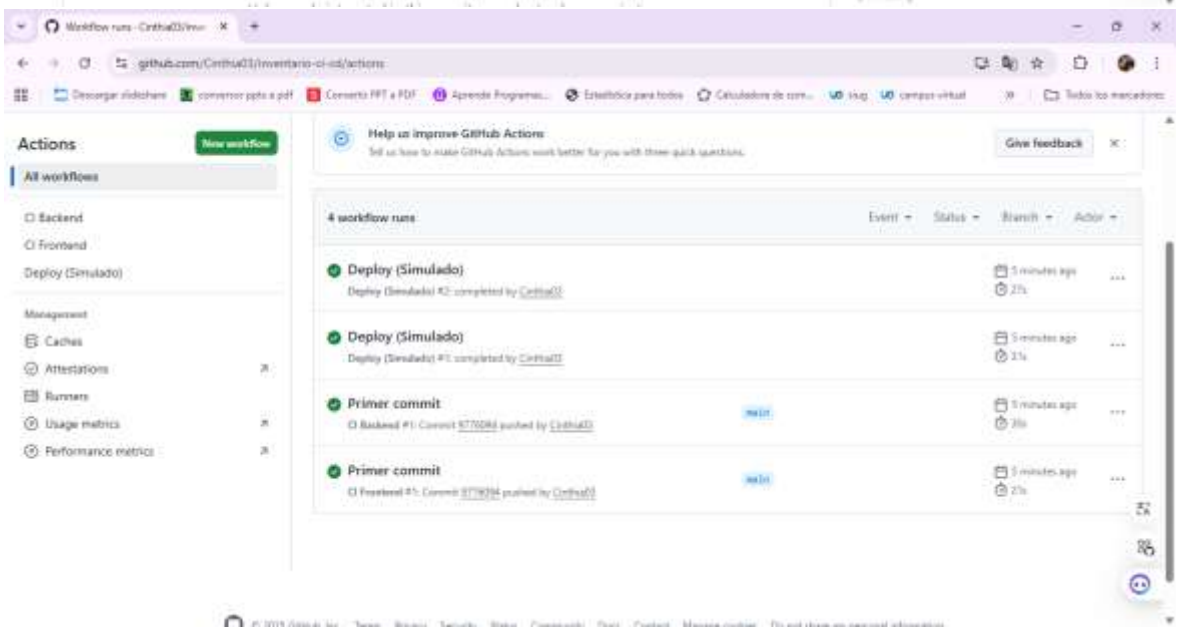


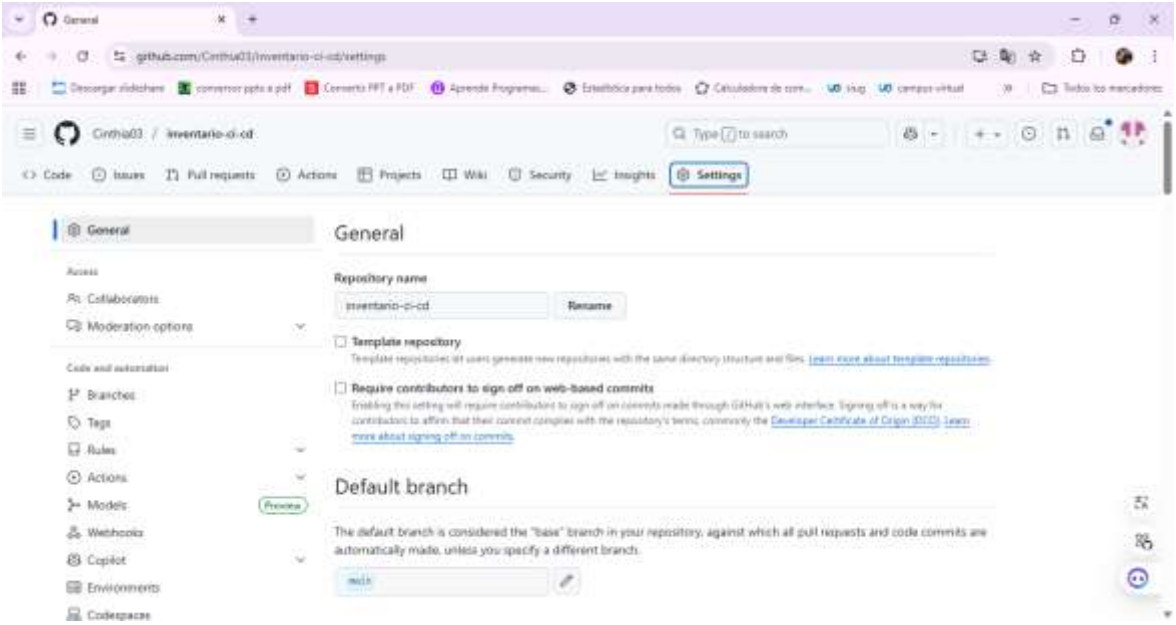


```
C:\Users\Cinth\Desktop\inventario-ci-cd>git commit -m "Primer commit"
[master (root-commit) d23996f] Primer commit
 5451 files changed, 525734 insertions(+)
 create mode 100644 .github/workflows/ci-backend.yml
 create mode 100644 .github/workflows/ci-frontend.yml
 create mode 100644 .github/workflows/deploy.yml
 create mode 100644 backend/jest.config.js
 create mode 100644 backend/node_modules/.bin/baseline-browser-mapping
 create mode 100644 backend/node_modules/.bin/baseline-browser-mapping.cmd
 create mode 100644 backend/node_modules/.bin/baseline-browser-mapping.ps1
 create mode 100644 backend/node_modules/.bin/browserslist
 create mode 100644 backend/node_modules/.bin/browserslist.cmd
 create mode 100644 backend/node_modules/.bin/browserslist.ps1
 create mode 100644 backend/node_modules/.bin/create-jest
 create mode 100644 backend/node_modules/.bin/create-jest.cmd
 create mode 100644 backend/node_modules/.bin/create-jest.ps1
 create mode 100644 backend/node_modules/.bin/esparse
 create mode 100644 backend/node_modules/.bin/esparse.cmd
 create mode 100644 backend/node_modules/.bin/esparse.ps1
 create mode 100644 backend/node_modules/.bin/esvalidate
 create mode 100644 backend/node_modules/.bin/esvalidate.cmd
 create mode 100644 backend/node_modules/.bin/esvalidate.ps1
 create mode 100644 backend/node_modules/.bin/import-local-fixture
 create mode 100644 backend/node_modules/.bin/import-local-fixture.cmd
 create mode 100644 backend/node_modules/.bin/import-local-fixture.ps1
 create mode 100644 backend/node_modules/.bin/jest
 create mode 100644 backend/node_modules/.bin/jest.cmd
 create mode 100644 backend/node_modules/.bin/jest.ps1
 create mode 100644 backend/node_modules/.bin/js-yaml
```

```
Símbolo del sistema
 create mode 100644 frontend/src/App.css
 create mode 100644 frontend/src/App.jsx
 create mode 100644 frontend/src/App.test.jsx
 create mode 100644 frontend/src/assets/react.svg
 create mode 100644 frontend/src/index.css
 create mode 100644 frontend/src/main.jsx
 create mode 100644 frontend/src/setupTests.js
 create mode 100644 frontend/vite.config.js

C:\Users\Cinth\Desktop\inventario-ci-cd>git branch
* master

C:\Users\Cinth\Desktop\inventario-ci-cd>git branch -m master main

C:\Users\Cinth\Desktop\inventario-ci-cd>git branch
* main

C:\Users\Cinth\Desktop\inventario-ci-cd>git push -u origin main
Enumerating objects: 5726, done.
Counting objects: 100% (5726/5726), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5501/5501), done.
Writing objects: 100% (5726/5726), 6.53 MiB | 1.21 MiB/s, done.
Total 5726 (delta 1212), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1212/1212), done.
To github.com:Cinthia03/inventario-ci-cd.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Cinth\Desktop\inventario-ci-cd>
```






General
github.com/Cinthia03/inventario-ci-cd/settings
Cinthia03 / inventario-ci-cd
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
General
Repository name
inventario-ci-cd
Rename
Template repository
Require contributors to sign off on web-based commits
Default branch
The default branch is considered the "base" branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.
main